1. Installation & Setup of React Native Environment

Option A: Using Expo (Recommended for beginners)

1. Install Node.js

   → https://nodejs.org

2. Install Expo CLI:
3. npm install -g expo-cli
4. Create project:
5. npx create-expo-app MyApp
6. cd MyApp
7. npm start
8. Run on phone using Expo Go

Option B: React Native CLI

1. Install Android Studio
2. Run:
3. npx react-native init MyApp
4. cd MyApp
5. npx react-native run-android

Practical 2: Basic React Native Components (Text, Image, Button)

```
import React from 'react';
import { View, Text, Image, Button } from 'react-native';

export default function App() {
  return (
    <View style={{ flex:1, justifyContent:'center', alignItems:'center' }}>
      <Text style={{ fontSize:24 }}>Hello React Native</Text>
      <Image
        source={{ uri:'https://reactnative.dev/img/tiny_logo.png' }}
        style={{ width:100, height:100, marginVertical:20 }}
      />
      <Button title="Click Me" onPress={() => alert('Button Pressed')} />
    </View>
  );
}
```

Practical 3: **UI Design using Core Components:** Design layout using View, Text, Image, and ScrollView components.

Index file:

```
import React from 'react';
import { Image, ScrollView, Text, View } from 'react-native';

export default function App() {
  return (
    <ScrollView>
      <View style={{ alignItems: 'center', marginTop: 50 }}>

        <Text style={{ fontSize: 24, fontWeight: 'bold' }}>
          Welcome to React Native
        </Text>

        <Image
           source={{ uri:
           'https://reactnative.dev/img/tiny_logo.png' }}
          style={{ width: 100, height: 100, margin: 20 }}
        />

        <Text>Student Name: Chetan</Text>
        <Text>Course: React Native</Text>

      </View>
    </ScrollView>
  );
}
```

Practical 4: **Styling using StyleSheet and Flexbox:** Apply styles using StyleSheet API for better code organization.

Index file:

```
import React from 'react';
import { StyleSheet, Text, View } from 'react-native';

export default function App() {
  return (
    <View style={styles.container}>

      <View style={styles.box1}>
        <Text style={styles.text}>Box 1</Text>
      </View>

      <View style={styles.box2}>
        <Text style={styles.text}>Box 2</Text>
      </View>

      <View style={styles.box3}>
        <Text style={styles.text}>Box 3</Text>
```

```
      </View>

    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    flexDirection: 'column',
    justifyContent: 'center',
    alignItems: 'center',
  },
  box1: {
    width: 100,
    height: 100,
    backgroundColor: 'red',
    margin: 10,
    justifyContent: 'center',
    alignItems: 'center',
  },
  box2: {
    width: 100,
    height: 100,
    backgroundColor: 'green',
    margin: 10,
    justifyContent: 'center',
    alignItems: 'center',
  },
  box3: {
    width: 100,
    height: 100,
    backgroundColor: 'blue',
    margin: 10,
    justifyContent: 'center',
    alignItems: 'center',
  },
  text: {
    color: 'white',
    fontWeight: 'bold',
  },
});
```

Practical 5: **User Input Form Creation:** Create forms using TextInput and Button components.

```
import React, { useState } from 'react';
import { Button, StyleSheet, Text, TextInput, View } from 'react-
native';

export default function App() {
  const [name, setName] = useState('');

  return (
```

```
    <View style={styles.container}>
      <Text style={styles.heading}>User Input Form</Text>

      <TextInput
        style={styles.input}
        placeholder="Enter your name"
        value={name}
        onChangeText={setName}
      />

      <Button title="Submit" onPress={() => alert(`Hello ${name}`)}
/>
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    justifyContent: 'center',
    padding: 20,
  },
  heading: {
    fontSize: 24,
    textAlign: 'center',
    marginBottom: 20,
  },
  input: {
    borderWidth: 1,
    padding: 10,
    marginBottom: 20,
  },
});
```

Practical 6: **Props and State in Functional Components:** Demonstrate passing data using props between components.

Product.tsx

```
import React from 'react';
import { StyleSheet, Text, View } from 'react-native';

export default function Product(props: any) {
    return (
        <View style={styles.card}>
            <Text style={styles.title}>Product: {props.name}</Text>
            <Text>Price: ₹{props.price}</Text>
        </View>
    );
}

const styles = StyleSheet.create({
    card: {
```

```
        padding: 20,
        marginBottom: 20,
        backgroundColor: '#f0f0f0',
        borderRadius: 10,
    },
    title: {
        fontSize: 18,
        fontWeight: 'bold',
    },
});
```

Index file

```
import Product from '@/components/Product';
import React, { useState } from 'react';
import { Button, StyleSheet, Text, View } from 'react-native';

export default function App() {
  const [cartCount, setCartCount] = useState(0);

  return (
    <View style={styles.container}>
      <Text style={styles.heading}>Shopping Cart Example</Text>

      <Product name="Laptop" price="50000" />

      <Text style={styles.cart}>Items in Cart: {cartCount}</Text>

      <Button
        title="Add to Cart"
        onPress={() => setCartCount(cartCount + 1)}
      />
    </View>
  );
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    justifyContent: 'center',
    padding: 20,
  },
  heading: {
    fontSize: 24,
    textAlign: 'center',
    marginBottom: 20,
  },
  cart: {
    fontSize: 20,
    marginBottom: 20,
  },
});
```

Practical 7: **Counter/Calculator App using State:** Develop a simple counter or calculator application.

```
import { useState } from 'react';
import { Button, StyleSheet, Text, View } from 'react-native';

export default function HomeScreen() {

  const [count, setCount] = useState(0);
  return (
    <View
      style={{marginTop: 50, padding: 16, flex: 1, flexDirection: 'column',
      gap:   10,justifyContent: 'center', alignItems: 'center',
    }}>

        <Text
            style={{ fontSize: 50, fontWeight: 'bold', textAlign:'center'
            }}>
        Welcome to Counter App!
        </Text>

      <Text style={{ fontSize: 40 }}>{count}</Text>

      <Button title="Increment" onPress={() => setCount(count + 1)} />

      <Button title="Decrement" onPress={() => setCount(count - 1)} />

      <Button title="Reset" onPress={() => setCount(0)} />
    </View >
  );
}
```